

THE TUNNELING DYNAMICS OF THE LOW TEMPERATURE H EXCHANGE REACTIONS

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A simple method for calculating of the tunneling linear hydrogen atom exchange reaction is proposed for heavy reactants. Its main feature is the competing of two small probabilities of a sufficiently close reactant approaching and of H transfer. The good accuracy and the validity of the whole tunnel dynamics picture are confirmed by the numerical results obtained by the classical S-matrix method modified for strongly exothermic reactions. The study of temperature dependences and isotopic effects of the linear H exchange reactions with a number of typical potential energy surfaces yields for the reaction barrier a value significantly higher than that given by simple one-dimensional barrier penetrability calculations.

1. Introduction

Strong experimental evidence of a tunneling mechanism has been obtained in the past years for a number of low-temperature reactions [1–13], particularly H abstractions from hydrocarbons by methyl radicals [5–10]



For this reaction the existence of a quantum-mechanical low-temperature rate constant limit was shown [10,11]. This effect was first discovered for the low-temperature polymerisation [13]. In this connection a theoretical study of the tunneling dynamics of H-transfer chemical reactions between heavy reactants is of great interest.

Even the studies of one-dimensional barrier penetrability [14–19] revealed the criterion for a dominant tunneling contribution to the reaction rate and permitted evaluation of the barrier height $U^\ddagger$ and width. A rough estimate of the penetrability is given by $\exp(-2\pi U^\ddagger / \hbar \Omega_i)$, where Ω_i is the reverse barrier frequency. However the concept of the static one-dimensional barrier as such is problematic, just as is the concept of Boltzmann distribution for the incident hydrogen atom, the vibration of which in the initial molecule is essentially quantal. Therefore the study of two- or more dimensional tunneling dynamics of the system is needed. Such attempts were made for

oscillator models of the reactions with use of the radiationless transition theory [20,21]. However these models only contain the parameters of the potential energy surface (PES) near the equilibrium positions of the reactants and products, and they are too crude for quantitative studies. Therefore simple methods are needed for calculating tunneling reaction rates with a realistic shape of PES. It is important that even the three-atom linear model of the reaction



allows one to choose the main coordinates for dynamic description and to realize the essential feature of the low-temperature chemical reactions – the competing of the small probability of the classical or tunnel approaching of the reactants to a sufficiently small distance R with the H-transfer tunneling factor which sharply decreases with R . It is important that for a collinear tunneling reaction an accurate classical S-matrix method (CSM) [22] is available and may serve for checking the accuracy of the approximated approaches.

The aim of the present work is to realize in terms of the linear reaction model the physical features of the tunneling dynamics of the reactive system (1), to provide simple methods for calculating the low-temperature reaction rate with the PES close to reality, and to study its temperature dependence and isotopic effect for typical PES.

Of interest is the reaction rate $W(T)$ which depends

on the characteristic frequency $\omega/2\pi$ of the reactant collisions and the mean dimensionless reaction probability $P(T)$ per collision

$$W(T) = (\omega/2\pi)P(T). \quad (3)$$

Assuming that the PES $U(R_{AH}, R_{HB})$ of the three-atom linear system (2) is known, consider the different approaches to calculating $W(T)$. We use a small parameter — the ratio of H atom and reactant masses in (2): $m_H/m_{A(B)} \ll 1$. For reaction (1), $m_B = 15 m_H$, $m_A \gg m_B$ (m_A may be less than the whole RH molecule mass if the approaching of the reacting parts of the molecule is due to its bending, rotation, etc.).

2. Classical motion of the reactants

In this approximation discussed earlier [23] the approaching of the reactants is considered to occur along the classical trajectory $R_{AB} = R(t)$ with energy E in the adiabatic potential

$$V_a(R) = U(R - r^e(R), r^e(R)) + \hbar[\Omega(R) - \Omega_{HB}]/2, \quad (4)$$

where $r^e(R)$, $\Omega(R)$ are the length and the stretching vibrational frequency of HB bond at $R_{AB} = R$. The transfer of H-atom is taken as instantaneous with probability $w(R)$ exponentially depending on R

$$w(R) = [\Omega(R)/2\pi] \exp[-2S(R, \epsilon(R))]. \quad (5)$$

Here S , the tunneling action integral,

$$S(R, \epsilon(R)) = \int_1^2 \{2m_H [U(R - r, r) - \epsilon(R)]\}^{1/2} dr, \quad (6)$$

is computed along the section $R_{AH} + R_{HB} = R$ of PES with the energy

$$\epsilon(R) = V_a(R) + \hbar\Omega_{HB}/2. \quad (7)$$

Even here the question arises about the role of the resonance of the hydrogen energy levels in both reaction channels at fixed $R_{AB} = R$. The treatment (5) corresponds to transition from the discrete level to the continuum and is justifiable in case of a sufficiently high exothermicity of the process and strong coupling of A—H vibration with other degrees of freedom such as bending vibrations and others. Then the probability

of the H transfer $w(R)$ depends only on the PES shape in the space of two coordinates and according to (5) it corresponds to transition from the initial quantum state $n_1 = 0$ to the arbitrary value of the vibrational quantum number n_2 in the product channel. This is the simplest way to avoid the difficulties caused by including other degrees of freedom and to express the rate constant in terms of the PES parameters. Note that a similar situation is observed for radiationless transitions in the molecules [24], where under certain conditions the probability does not depend on the broadening mechanism which allows the transition. An alternative mechanism of H transfer reaction essentially dependent on the resonance of the hydrogen levels was studied in ref. [25].

During the reactant's motion along the trajectory $R(t)$ with energy E and collision frequency $\omega/2\pi$ the reaction rate is controlled by the turning point $R_t = \min R(t)$ where $V_a(R_t) = E$, and the main contribution to the Boltzmann averaged rate $W(T)$ is given by the energy region around the optimal value $E(T)$ at which

$$\beta = \frac{1}{kT} = 2 \frac{dS(R, \epsilon(R))/dR}{dV_a/dR}. \quad (8)$$

Thus we have

$$W_{cl}(T) = \frac{\omega}{2\pi} A_0 \left(2\pi \frac{dE(T)}{dkT} \right)^{1/2} e^{-\beta E - 2S(R, \epsilon)}, \quad (9)$$

$$A_0 = \frac{\Omega}{2\pi} \left(\frac{dS}{dR} \frac{dV_a}{dR} \frac{1}{\pi M} \right)^{-1/2} \quad (10)$$

where $M = m_A m_{HB}/(m_A + m_{HB})$ is the reduced mass of the reactants. The function $E(T)$ represents the temperature dependent activation energy of the linear reaction.

3. Account for the tunneling approaching of reactants (exothermic reaction)

In reality the H transfer may also occur at the classically inaccessible tunneling distance $R < R_t$ with the probability

$$w(R) \psi_E^2(R) dR = w(R) \frac{\omega}{2\pi} \frac{M}{P(R)} \exp \left(-2 \int_{R_t}^R P(R') dR' \right) dR,$$

$$P(R, E) = [2M(V_a(R) - E)]^{1/2}, \quad (11)$$

where E is the relative energy of the reactant motion. Now the H-transfer takes place mainly at such values R^* at which the function of full action integral

$$2\phi(R) = 2S(R, \epsilon(R)) + 2\kappa\hbar^{-1} \int_{R_t}^R P(R', E) dR', \quad (12)$$

has a minimum. Here $\kappa = 1$ for exothermic reaction. This gives

$$d\phi(R^*)/dR = G(R^*) - \kappa\hbar^{-1}P(R^*, E) = 0 \quad (13a)$$

$$G(R^*) = dS(R, \epsilon(R))/dR. \quad (13b)$$

It means from (13) that the given value R^* of the H-transfer distance corresponds to the energy

$$E = V_a(R^*) - \hbar^2 G^2/2M. \quad (14)$$

Then the thermally averaged reaction rate will be

$$W(T) = (\omega/2\pi) A_R A_T \exp[-2\phi(R^*(E)) - \beta E], \quad (15)$$

$$A_T = (2\pi dE/dkT)^{1/2};$$

$$A_R = \frac{\Omega}{2} \left[\left(G \frac{dV_a}{dR} - \frac{dG}{dR} \frac{\hbar^2 G}{2M} \right) \frac{\pi}{M} \right]^{-1/2}. \quad (16)$$

Here the temperature dependent activation energy $E(T)$ is defined by the equation

$$2d\phi/dE = 2\tau(E)/\hbar = \beta, \quad (17)$$

which connects the reverse temperature $\beta = 1/kT$ with the imaginary time $-i\tau$ of the reactant tunneling motion

$$\tau(E) = \int_{R_t}^{R^*} dRM/P(R, E). \quad (18)$$

The result (15) remains correct even at $R^* \rightarrow R_t$ when $\psi_E^2(R)$ in eq. (11) needs the Airy representation. Particularly at $M/m_H \rightarrow \infty$, the rate constant $W(T)$ from eqs. (13) and (17) only slightly differs from the classical one defined by eqs. (8) and (9):

$$W(T) = W_{cl} \exp \left[\frac{-\hbar^2}{3M\kappa^2} \left(\frac{dS}{dR} \right)^3 \frac{dV}{dR} \right]. \quad (19)$$

4. Symmetrical reaction $A + HA \rightarrow AH + A$

Unlike the exothermic reaction, the tunneling ap-

proach of the reactants in the symmetric reaction is followed by the tunneling separation of the products, and the reaction rate $W_{sym}(T)$ must be obtained as the rate of the transition from the relative energy state E_1 of reactant to the final states E_2

$$W(E) = \frac{2\pi}{\hbar} \frac{1}{\hbar\omega} \left| \int \psi_{E_1}^*(R) V_{12}(R) \psi_{E_2}(R) dR \right|^2, \quad (20)$$

due to the interaction $V_{12}(R)$ which is equal to the tunneling splitting of the resonant hydrogen levels at fixed R [26]

$$V_{12}(R) = (\hbar\Omega/2\pi) \exp[-S(R, \epsilon(R))].$$

The wavefunctions in (20) are unit normalized as in (11), though eq. (20) as usual corresponds to transition to the continuum. The integral in (20) contains the exponent with argument

$$\phi(R) = S(R, \epsilon) + 2 \int_{R_t}^R P(R', E) dR',$$

and its point of steepest descent is defined by eq. (13) with $\kappa = 2$. As a result the reaction rate will be given by eq. (15), where $R^*(E)$, $E(T)$ and ϕ are determined by eqs. (13), (17) and (12) with $\kappa = 2$ and the pre-exponential factor A_R must be replaced by

$$A_R^{sym} = \frac{\Omega^2}{2\pi} M \left(G \frac{dV_a}{dR} - \frac{dG}{dR} \frac{\hbar^2}{8M} G^2 \right)^{-1}. \quad (21)$$

It is easy to verify that at $M/m_H \rightarrow \infty$ the near classical results (19) with $\kappa = 2$ may be used with replacement of A_R by A_R^{sym} in eq. (9).

5. The quantum-mechanical low-temperature rate constant limit

The behaviour of $W(T)$ at $T \rightarrow 0$ depends significantly on the shape of the PES at large distances R . For a reaction in a solid matrix the existence of a potential minimum V_0 at some equilibrium distance $R_{AB} = R_0$ with corresponding reactant vibration frequency ω_0 is typical. In order to find the rate constant's quantum-mechanical limit $W_0 = \lim_{T \rightarrow 0} W(T)$, the averaging over the H-transfer distance R and over the classical reactant energy must be replaced by the averaging $W(R)$ over the quantal temperature partition function $\rho_\beta(R)$ in a potential $V_a(R)$. In the tun-

neling region R it is

$$\rho_\beta(R) = (2\pi D)^{1/2} Z^{-1} \exp[-F(R, R', \beta)]|_{R'=R}, \quad (22)$$

$$F(R, R', \beta) = \int_{R_t}^R P(R, E) dR + \int_{R_t}^{R'} P(R, E) dR + \beta(E - V_0), \quad (23)$$

$$Z = \exp(-\beta\hbar\omega/2) [1 - \exp(-\beta\hbar\omega)]^{-1},$$

$$D(R) = \frac{\partial^2 F(R, R', \beta)}{\partial R \partial R'} \Big|_{R=R'} = -\frac{1}{\hbar} \frac{\partial E(R, R, \beta)}{\partial \beta} \frac{M^2}{P^2(R, E)} \quad (24)$$

Here the energy $E = E(R, R, \beta)$ is defined by eqs. (17) and (18). Then the reaction rate

$$W(T) = \int \rho_\beta(R) w(R) dR$$

is given by the same expression (15) with replacement of A_T by its quantal value

$$A_T^q = [(1 - e^{-\beta\hbar\omega})/\hbar\omega] (2\pi e^{\beta\hbar\omega} dE/d\beta)^{1/2} \quad (25)$$

It coincides with A_T at $\beta\hbar\omega \ll 1$.

Now with account for the parabolic behaviour

$V_a(R)$ near R_0 eq. (16) gives if $E' = E - V_0 \rightarrow 0$

$$\beta(E') = (\hbar\omega)^{-1} \ln(\hbar\omega/E') + \beta_0 + \gamma E' + O(E'^2), \quad (26)$$

or

$$\ln \frac{W(T)}{W_0} = \int_0^{E'(\beta)} \frac{d\beta(E')}{dE} (E - V_0) dE \approx e^{(\beta - \beta_0)\hbar\omega}. \quad (27)$$

Here the quantum-mechanical rate constant limit is

$$W_0 = (2\pi)^{1/2} e^{\beta_0\hbar\omega/2} A(R_0^*) \exp[-2\phi(R_0^*)].$$

where ϕ , A_R and the H-transfer distance $R_0^* = R^*(E = V_0)$ are defined by eqs. (12), (16) and (13). The characteristic temperature β_0 of the rate tending to the limit is connected with double imaginary time of the reactant motion from $R = R_0 - 2^{1/2}X_R$ to R_0^* with $E = V_0(X_R = (\hbar/M\omega_0)^{1/2})$ is the zero reactant vibration amplitude). If the exponential repulsive potential $V_a(R) \approx A \exp(-\alpha R)$ at large R occurs, then the approach $W(T) \rightarrow W_0$ becomes $\ln(W(T)/W_0) \approx 2M/\hbar^2 \alpha^2 \beta$ as $T \rightarrow 0$.

6. The tunneling trajectories in the classical S-matrix (CSM) method [22]

We shall first describe the rigorous CSM method proposed by Miller et al. [22] and its modification for exothermic reactions. The accuracy of all approximate methods will be checked further by comparison of their results with the numerical ones obtained by CSM method.

In CSM the dimensionless tunneling reaction probability $P_{n_1 n_2}(T)$ per reactant collision from its ground vibrational state $n_1 = 0$ to the final state n_2 of the product AH is connected with the characteristics of the complex-valued classical trajectory of the system

$$P_{n_1 n_2}(T) = A_0 A_T \exp[-2\text{Im}\phi_{n_1 n_2}(E) - \beta E], \quad (28)$$

$$A_0 = [2\pi dn_2(\varphi_1 n_1)/d\varphi_1]^{-1},$$

$$A_T = [2\pi dE/dkT]^{1/2}. \quad (29)$$

$\phi_{n_1 n_2}$ is the classical action integral along this trajectory defined by the double-ended boundary condition at sufficiently large distances $R_a(t_1)$, $R_b(t_2)$:

$$n_1^a(t_1) = n_1 = 0, \quad P_a(t_1) = -(2M_a E)^{1/2} \quad (30)$$

$$n_2^b(t_2) = n_2, \quad P_b(t_2) = [2M_b(E + Q)]^{1/2}. \quad (31)$$

Here $n^{a(b)}$, $\varphi^{a(b)}$ are the vibrational actions and phases, $P_a(b)$, $R_a(b)$, $M_a(b)E$, $E + Q$ are the momenta, coordinates, masses and energies of the relative motion in the reactant (a) and product (b) channels. The activation energy $E = E(\beta)$ in (28) is defined by the equation

$$2\tau_\Sigma(E) = -2\text{Im}(t_2 - t_1) = \hbar\beta, \quad (32)$$

where τ_Σ is the imaginary part of the full time increment along the trajectory. For linear models of the exothermic reaction, some modifications of the usual CSM treatment are needed. Indeed its rate is the sum of the probabilities for all final vibrational states. For the linear reaction (2) with heavy A and B fragments and large H-bond quantum, the usual CSM probabilities depend sharply on a fulfillment of the resonance of hydrogen vibrational levels in both channels. This reflects the fact that the small ratio m_H/M makes difficult the energy exchange between the vibrational and relative motion and indicates to the restrictions of the linear model. The latter is removed if the rearrangement of other degrees of freedom in the course

of the reaction is taken into account (see discussion in section 2). Assume that the dissipation of excess energy into such degrees of freedom is sufficiently fast and does not limit the process rate. Then the reaction rate may be defined as the transition probability to the most probable (generally noninteger) final quantum number n_2^* :

$$\bar{P} = \sum_{n_2} P_{0n_2} \rightarrow \int P_{0n_2} dn_2 = BP_{n_0 n_2^*},$$

where n_2^* determines the minimum of the imaginary part of the action integral:

$$\left. \frac{d \operatorname{Im} \phi_{0n_2}}{dn_2} \right|_{n_2^*} = \operatorname{Im} \varphi_2(n_1 \varphi_1) = 0; \quad \operatorname{Im} n_2^* = 0. \quad (33)$$

Thus the exothermic reaction probability is considered to be

$$P(T) = BA_0 A_T \exp(-2 \operatorname{Im} \phi_{0n_2^*} - \beta E),$$

$$B = (\pi^{-1} d \operatorname{Im} \varphi_2 / dn_2)^{1/2}, \quad (34)$$

where A_0 and A_T are given by (28) and all quantities are calculated along the trajectory with the boundary conditions (30) and new ones (33) requiring the real values for $n_2(\varphi_1, n_1)$, $\varphi_2(n_1, \varphi_1)$.

Our calculations of the symmetric reaction probability have been carried out with aid of the double-ended procedure described earlier in detail [22,27]. For the exothermic reactions the finding of the classical trajectories with boundary conditions [eqs. (30) and (33)] was based on the direct integration procedure into product channel [22]. The initial phase φ_1 for the first value of the energy E for given PES has been obtained by scanning over the reaction exoergicity starting from symmetric PES, because the iterational procedure with the direct integration converges only near the needed value φ_1 . The CSM results for LEPS-type PES are discussed below.

7. The approximate tunneling reactive trajectory

The approximate method of constructing such trajectories is based on the smallness of the ratio m_H/M . Consider three parts of the reactive tunneling trajectory: (1) the classical and then tunneling approaching of the reactants, (2) the tunneling H-transfer from one channel to another accompanied by the imaginary in-

crement $\Delta t = -i\tau_H$ and small displacement of the reactants, and (3) the drawing apart of the products. Take into account that the period $2\pi/\Omega_{CH}$ of the C-H vibrations, just as the time $\tau_H = \pi/\Omega_i$ of the H-transfer, are essentially less than the time of heavy reactant motion. Therefore the stage of the classical reactants approaching the turning point R_t and then the tunneling up to the time $t'_1 = t_t - i\tau_1$, where $\operatorname{Im} \varphi_a(t'_1) = 0$ (the turning point of the vibrational motion) appears to be adiabatic relative to the vibrations, so that at $t = t'_1$ we have

$$P_a(R_1) = -iP(R_1, E) = -i[2M(V_a(R_1) - E)]^{1/2};$$

$$\epsilon(n_a(t_1)) = \hbar\Omega(R)/2, \quad (35)$$

and the distance $R_1 = R_{AB}(t'_2)$ is connected with the imaginary part of the initial phase φ_1 :

$$\operatorname{Im} \varphi_1 = \int_{R_t}^{R_1} \Omega(R) M dr / P(R, E) \approx \Omega \tau_1. \quad (36)$$

The further time increment in the negative imaginary direction is accompanied by rapid H-transfer to the product channel. In this second region between two vibrational turning points t'_1 and t'_2 (1 and 2) in both channels, the trajectory may be found by expansion in the small parameter m_H/M . The zero order gives (see appendix A) $R^0(\tau) = \text{const.} = R_1$ and

$$P^{(0)}(t'_2) = iP_2 = P(t'_1) - i\hbar\tilde{G}, \quad (37)$$

$$\tilde{G} = \partial S(R, \epsilon(R)) / \partial R$$

$$= \frac{1}{\hbar} \int_1^2 \frac{\partial U}{\partial R} \{2[U(R-r, r) - \epsilon(R)]/m_H\}^{-1/2} dr, \quad (38)$$

where $\epsilon(R) = V_a(R) + \hbar\Omega_{HB}/2$ and S is given by eq. (6). For a symmetric reaction the condition following from the trajectory symmetry is $P_2 = -P_1$. At the same time for the exothermic reaction the boundary condition (33) requires that $P(t'_2) = 0$. For both cases, the equation determining the zero order H-transfer distance R_1 is

$$\{2M[V_a(R_1) - E]\}^{1/2} = \kappa^{-1} \tilde{G} = \kappa^{-1} \partial S(R, \epsilon) / \partial R, \quad (39)$$

where $\kappa = 1$ or $\kappa = 2$ stand for exothermic or symmetric reactions, respectively. Then the imaginary part of the action integral along the whole trajectory becomes

$$\phi(R_1) = \phi^{(0)}(R_1) + \phi^{(1)}(R_1). \quad (40)$$

Here $\phi^{(0)}$ exactly coincides with eq. (12) where R_1 is determined however by eq. (39) instead of eq. (13).

As a result the activation energy E corresponds to the temperature according to

$$\beta = 2 \frac{d\phi}{dE} = 2 \left(\frac{\partial \phi}{\partial E} + \frac{\partial \phi}{\partial \epsilon} \frac{d\epsilon}{dR} \frac{dR}{dE} \right),$$

or

$$\hbar\beta/2 = \tau(E, R_1) + \tau_H + \delta\tau_H \approx \tau(E, R_1) + \tau_H, \quad (41)$$

where $\tau(E, R)$ is the time for the tunneling motion of the reactants (products) defined by (18), and τ_H is the time of the tunneling H-transfer

$$\begin{aligned} \tau_H &= \hbar \partial S(R, \epsilon) / \partial \epsilon \\ &= \int_1^2 dr \{ 2 [U(R-r, r) - \epsilon(R)] / m_H \}^{-1/2}. \end{aligned} \quad (42)$$

The correction $\delta\tau_H$ of the next order in m_H/M in (41) is given by (A.7). It appeared to be $\delta\tau_H \ll \tau_H$ in the calculations made, and it might be neglected. The next order in the expansion in m_H/M allows obtaining the action correction $\phi^{(1)}$ and the reactant distance variation δR in a time τ_H

$$\phi^{(1)} = \frac{1}{2M\hbar} \int_1^2 \{ 2P_1 F(\tau) - F^2(\tau) \} d\tau, \quad (43)$$

$$\delta R = M^{-1} \int_1^2 P(\tau) d\tau; \quad P(\tau) = P_1 - F(\tau), \quad (44)$$

$$F(\tau) = \int_1^\tau \frac{\partial U}{\partial R} d\tau;$$

$$d\tau = dr \{ 2 [U(R-r, r) - \epsilon(R)] / m_H \}^{-1/2}. \quad (45)$$

It is shown in appendix A that to the highest order in M/m_H , the pre-exponential factor A_R in the rate constant expression (15) with R_1 from eq. (39) coincides with those (16) or (21) obtained previously, where G is replaced by $\tilde{G}$ given by (38). A more exact value of A_R is given by (A.13) or (A.10).

Thus the procedure of finding the rate constant consists of computation of the integrals (6), (38), (42)–(45) between the vibrational turning points 1 and 2 in both channels along the PES section $R_{AH} + R_{HB} = R$.

Then eq. (14) with $G \rightarrow \tilde{G}$ gives the reactant energy E and therefore allows finding the reactant tunneling time connected with temperature according to eq. (41).

8. Comparison of the CSM and approximate results

Such comparison allows one not only to evaluate the accuracy of the approximate method, but also to appreciate the meaning of the complex-valued CSM trajectory of the system. In dynamical CSM calculations the standard LEPS-type [28,29] PES I.1–I.4 from table 1 have been used with the barrier height values 8–14 kcal/mol expected according to refs. [30–33]. The parameters of this PES in the same designations as in ref. [27] are $R_i^e = 1.0934, 1.54, 1.0934$ Å, $\alpha_i^s = \alpha_i^T = 1.78, 1.54, 1.78$ Å⁻¹, $i = 1, 2, 3$, $D_1 = 109$, $D_2 = 84.4$ kcal/mol. The varying dissociation energy D_1 , the Sato parameters k_i and the resulting saddle point characteristics of the PES used are shown in table 1. The values $m_A = m_B = 15 m_H$ the smallest possible values for reaction (1) were taken to evaluate the maximal quantum-mechanical effects from the reactant motion.

Fig. 1 represents the resulting probabilities $P(T)$ of the tunneling reaction per collision as functions of the inverse temperature. The classical approximation is seen to significantly decrease the probability. Accounting for the reactants tunneling motion with the assumption of instantaneous H-transfer ($\tau_H = 0$) gives overestimated probability values. At the same time, the calculations according to eqs. (6), (38), (42)–(44), (A.10) and (A.13) with first order in m_H/M corrections (points in fig. 1) give surprisingly good agreement with CSM results (full line). The using of zero order expressions (16) and (21) with $G \rightarrow \tilde{G}$ instead of (A.13) and (A.10) for the pre-exponential factor gives the values greater than (A.13) and (A.10) or the CSM ones by a factor about 1.5. For the deuterium (D) transfer reaction the same calculations give a difference about $\Delta \log P \approx 0.5$ compared to the CSM probabilities because here the parameter m_D/M is ≈ 0.3 , and $\phi^{(1)}/\phi^{(0)}$ becomes large. This leads to three times too small values for the isotopic rate constant ratios $\eta = P_H(T)/P_D(T)$, which are about 10^2 – $10^{3.5}$ at $T = 70$ K for the PES studied.

It is important that the proposed approximate method allows prediction not only of probabilities, but

Table 1
Saddle point PES parameters for various D_1

	$k_i = 0.1, 0, 0.1$				$k_i = 0.03, 0, 0.03$		
no.	I.1	I.2	I.3	I.4	II.1	II.2	II.3
D_1 (kcal/mol)	95	100	105	109	95	100	105
$U^\ddagger$ (kcal/mol)	7.99	9.77	11.88	13.83	13.02	15.15	17.54
R_1 (Å)	1.226	1.255	1.291	1.324	1.263	1.290	1.319
R_2 (Å)	1.481	1.419	1.364	1.324	1.466	1.419	1.377
ω_i (cm ⁻¹)	1359	1598	1738	1772	1809	1940	2019

also of such characteristic of the tunneling CSM trajectories as the H transfer distance R_1 or its imaginary initial phase values. In fig. 2 some CSM trajectories (the real parts of their coordinates) and the corresponding approximate predictions for the H-transfer distance are presented. Fig. 3 shows comparison of the CSM functions $\text{Im } \varphi_1$ of the reverse temperature with predicted dependence

$$\text{Im } \varphi_1 = \kappa^{-1} \Omega (\hbar\beta/2 - \tau_H), \quad (46)$$

following from eq. (36) ($\kappa = 1$ or 2 for exothermic or symmetrical reaction).

A linear relation like (46) between $\text{Im } \varphi_{1(2)}$ and β is well fulfilled for the $\text{H}_2 + \text{H}$ reaction, as checked by the CSM calculations of this reaction [27]. In this connection it is interesting to note the suggestion [34] of

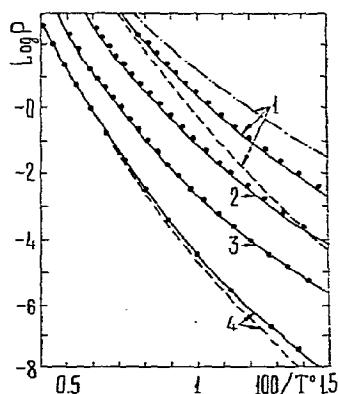


Fig. 1. Comparison of the CSM reaction probabilities $P(T)$ (full line) with those calculated in zero and first order in m_H/M (points) for the exothermic PES I.1–I.3 and for symmetric PES I.4 from table 1. The dashed and dashed-dotted lines refer to classical approximation results and the instantaneous H transfer approximation result respectively for PES I.1, I.4.

evaluating the total tunneling factor: by the action integral along the path removed relative to the reaction path by the zero amplitude of the orthogonal vibrations. However as may be seen in fig. 2 the tunneling path of the reaction (2) with $m_A, m_B \gg m_H$ differs significantly from that suggested in ref. [34].

It is to be noted also that unlike the one-dimensional tunneling trajectories, the two-dimensional ones cannot be made real by transformations $P_j \rightarrow iP_j$, $\Delta r \rightarrow i\Delta r$ in the whole tunneling motion region. This is connected with the existence of two different time scales (of H and A B-motions) and with the necessity of accounting for the C–H vibration zero point energy which cannot be neglected compared to the barrier height. It is interesting also to compare the CSM tunneling trajectory with that in the first steepest descent approximation of the radiationless theory models of the reaction [20,21]. In the latter the trajectory connects two turning points of the total motion (translation and vibration) – in both channels: $P_{R_i}(t_i) = 0$, $P_{r_i}(t_i) = 0$, $i = 1, 2$ provided $t_2 - t_1 = -i\hbar\beta/2$, so that the substitution $P'_k \rightarrow iP_k$, $t \rightarrow i\tau$ makes it entirely real. It is easy to show that when $\beta \rightarrow \infty$ ($T \rightarrow 0$) energy E_t of such a tunneling trajectory tends to zero (instead of $E_t \rightarrow \hbar\Omega/2$ for the CSM trajectory) as is consistent with the same tunneling time $\beta/2$ for both fast (r_H) and slow (R_{AB}) degrees of freedom. This leads to an error in the reaction probability which may be evaluated as

$$\Delta \ln P \approx 2 \frac{\partial S(R, \epsilon)}{\partial \epsilon} \frac{\hbar\Omega}{2} = 2\tau_H\Omega,$$

and gives two orders of magnitude.

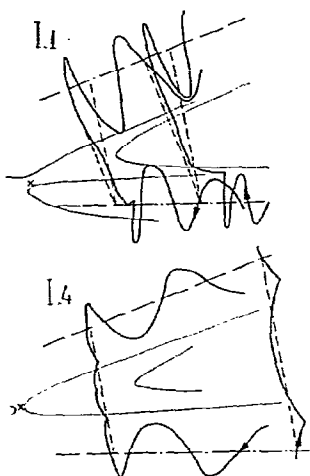


Fig. 2. The CSM trajectories (the real parts of their coordinates) in the usual R_{HA} , R_{HB} plot for the exothermic PES 1.1 with $E = 4.6$ and 2.4 kcal/mol and for symmetric PES 1.4 with energies 8 and 3.6 kcal/mol. The dashed lines refer to the corresponding H-transfer distances $R_{AB} = R_1^0(E)$ or connect the points $R_{AB} = R_1^0(E)$ in the initial channel with $R_{AB} = R_1^0 + \delta R^{(1)}$ in final one (for PES 1.4 $\delta R^{(1)} = 0$). The crosses designate the saddle points of PES and dashed-dotted lines defined by $R_{AH}(HB) = R_i^0$.

9. Study of the temperature dependence and the isotope effect

For this purpose use the approximate method based on eq. (6), (38), (42)–(44) providing sufficient accuracy. Correct models of the solid matrix reaction must include the minimum $V_0 = V_a(R_0)$ in the reactant potential at a distance R_0 close to the sum of the van der Waals radii of H and C atom: $R_0 = R_{CH} + R_C + R_H = 3.8$ – 3.9 Å [35]. To describe it the triplet energies E_{T_1} and E_{T_3} in the standard LEPS expression were replaced by

$$E_{T_i} = D_T \{ [\exp(-\alpha_i \Delta R_i - f_i(R_i)) + 1]^2 - 1 \}, \quad (47)$$

$$f_i(R_i) = e^{\alpha_i(R_i - d)} - e^{\alpha_i(R_i^\ddagger - d)}, \quad i = 1, 3$$

(the usual E_T^i corresponds to $f_i(R) = 0$). This provides a sufficiently rapid decrease $E_T^i(R)$ at large R and practically does not change the saddle point characteristics. In table 2 the approximate PES parameters near $R_{AB} = R_0$ are given for two values of d (they have very slight dependence on D_1).

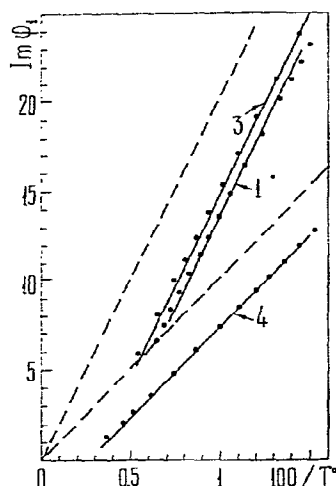


Fig. 3. Imaginary parts of initial phases as functions of reverse temperature for PES 1.1–1.4. Full lines refer to the CSM results, the points are those predicted by eq. (45), the dashed lines correspond to the instantaneous H transfer ($\tau_H = 0$).

Calculations show that the low-temperature activation energy depends only slightly on the reaction exothermicity, but varies sharply with variation of the reactant equilibrium distance $R_0(d)$. Fig. 4 represents the temperature dependences of $P(T)$ for PES 11.1 from table 1 with $D_1 = 95$ kcal/mol, as an example of tending to the quantum-mechanical limit. For $D_1 = 100$ and 105 kcal/mol the functions $P(T)$ at low temperature are similar to those presented in fig. 4, but less by about 1.5 and 3 orders of magnitude. Table 2 lists the values of the low temperature rate constant limit W_0 and of the maximal quantity of an isotope ratio $\eta_0 = W_0(H)/W_0(D)$. Since the modification (46) in LEPS PES leads to some increasing in the H transfer barrier at large distance, the study of the temperature dependence of $P(T)$ has been carried out also for a PES where the reactant potential minimum is made by the simple addition of term $U(R_{AB}) = A \exp(\alpha R_{AB})$ instead of (47). However the T -dependence of $P(T)$ at low T is similar to that calculated with modified LEPS (47) with small equilibrium values R_0 , ω_0 although the absolute values of $P(T)$ depend on total reaction barrier $V^\ddagger - V_0$.

According to the rough estimate given in appendix B the three-dimensional rate constant W_{3-D} seems to be of the same order of magnitude as that for the

Table 2

Limiting probabilities P_0 , rate constant W_0 , (sc^{-1}), and isotope ratios $\eta_0 = W_0(\text{H})/W_0(\text{D})$ of linear reactions with PES parameters from table 1, $m_B = 40 m_H$ and various values d , ($\text{\AA}$). Equilibrium reactant characteristics $U_0(\text{kcal/mol})$, $R_0(\text{\AA})$, $\omega_0(\text{cm}^{-1})$, and the total barrier height $U^\ddagger - U_0(\text{kcal/mol})$ are also presented

	$d = 2.5$			$d = 2.7$			$d = 2.5$			$d = 2.7$		
PES number	I.1	I.2	I.3	I.1	I.2	I.3	II.1	II.2	II.3	II.1	II.2	II.3
$U^\ddagger - U_0$	9.76	11.56	13.68	9.29	11.08	13.20	14.63	16.77	19.17	14.19	16.68	18.72
$-\log P_0$	10.3	11.5	12.6	11.9	13.3	14.6	13.76	14.9	16.1	15.8	17.2	18.3
W_0^a	1.4 ₂	8.8	0.70	2.9	1.15 ₋₁	5.7 ₋₃	7.2 ₋₂	3.6 ₋₃	2.3 ₋₄	3.8 ₋₄	1.5 ₋₅	1.2 ₋₆
$\log \eta_0$	3.3	3.7	4.2	3.3	3.6	4.2	4.3	4.8	5.3	4.4	4.9	5.4
	$U_0 = -1.77, R_0 = 3.75, \omega_0 = 87$			$U_0 = -1.13, R_0 = 3.89, \omega_0 = 72$			$U_0 = -1.61, R_0 = 3.83, \omega_0 = 89$			$U_0 = -1.17, R_0 = 3.99, \omega_0 = 74$		

a) The subscripted numbers are powers of ten by which the preceding number must be multiplied.

linear reaction W_{1-D} . With this account the experimental value $W_0 \approx 3 \times 10^{-4} \text{ s}^{-1}$ for the H atom abstraction by the CH_3 radical from methanol for the isotope effect $\eta_0 \geq 2.8 \times 10^4$ [10,12] and the small activation energy at low temperature corresponds to the small equilibrium distances $R_0 \approx 3.8 \text{ \AA}^2$ between the reactants.

The linear reaction calculations change our concept of the tunneling dynamics and allow us to make an important conclusion about the high value $> 15 \text{ kcal/mol}$ of the reaction barrier height compared with the typical parameters of one-dimensional gaussian reaction barrier $V_0 = 8.4 \text{ kcal/mol}$, $a = 0.687 \text{ \AA}^{-1}$, $\hbar\Omega_i = 610 \text{ cm}^{-1}$, $\beta' = (2\pi V_0/\hbar\Omega_i) = 28.5$. For the latter a rough estimation gives for the barrier penetrability $P \leq \exp(-\beta') = 10^{-13}$. But the same order of the

linear reaction probability magnitude is obtained for PES II.1 from table 1 with $d = 2.5$, where the H-transfer for $m_c = 32 m_H$ at $T = 32 \text{ K}$ takes place at $R = 3.3 \text{ \AA}$ under the barrier

$$V_0(R) = \max V(R-r, r) - V_a(R) - \hbar\Omega/2 = 20.6 \text{ kcal/mol}$$

with $\hbar\Omega_i \approx 1800 \text{ cm}^{-1}$ (the typical values for such reaction). As a result we have

$$\log P_0 = 2 \left(S + \int_{R_t}^{R_0^*} P dR \right) \log e$$

$$= 2(11.3 + 4.8) 0.43 = 13.8.$$

Thus the linear reaction treatment is undoubtedly useful though it does not allow to study how does the changing of the adjacent CH bond angles in the course of the H-transfer influence the reaction rate, and to realize the mechanism of the exothermicity dissipation, etc.

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Appendix A. Low orders of the expansion in m_H/M

In the region between the points 1' and 2' of the

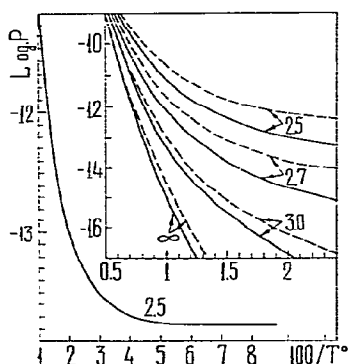


Fig. 4. Reaction probabilities $P(T)$ for sets of modified LEPS PES with parameters II.1 from table 1 and various values of parameter d written near the curves. The full and dashed lines correspond to $m_B = 40 m_H$ and $m_B = 15 m_H$, respectively.

vibrational turning points in both channels, the classical equations in the variables $R = R_{AB}$, $r = r_{HB}$, $\tilde{P}_R = iP_R$, $\tilde{P}_r = iP_r$ are

$$\dot{r} = \frac{1}{m_H} \tilde{P}_r; \quad \dot{\tilde{P}}_r = -\frac{\partial \tilde{U}}{\partial r}; \quad \dot{R} = \frac{1}{M} \tilde{P}_R; \quad \dot{\tilde{P}}_R = -\frac{\partial \tilde{U}}{\partial R}. \quad (\text{A.1})$$

with $\tilde{\epsilon} = -\epsilon$, $\tilde{U} = -U$ and the derivatives over $\tau = it$. The boundary conditions at $t = t'_1$ and $t = t'_2$ are

$$\tilde{P}_r(1) = 0; \quad \tilde{P}_R(1) = iP_1 = -[2M[V_a(R_1) - E]]^{1/2}, \quad (\text{A.2})$$

$$\tilde{P}_r(2) = 0; \quad \tilde{P}_R(2) = (1 - \kappa)P_R(1),$$

$\kappa = 2, 1$ for symmetry or exothermic reactions. In zero approximation $R^{(0)}(\tau) = \text{const.}$ and for $P(\tau)$ and $S^{(0)}$ we have eqs. (44) and (6). In next order we have

$$\dot{\tilde{P}}_r = \frac{d}{d\tau} \left(\frac{\dot{r}^2}{2m_H} \right) = -\frac{d\tilde{U}}{d\tau} - \frac{\partial \tilde{U}}{\partial R} \frac{dR^{(1)}(\tau)}{d\tau}, \quad (\text{A.3})$$

where $\tilde{U} = \tilde{U}(R^{(0)} - R^{(1)}(r) - r, r)$. This gives

$$d\tau = dr [2\{\tilde{\epsilon} - \tilde{U} - f^{(1)}(r)\}/m_H]^{-1/2}, \quad (\text{A.4})$$

$S^{(0)} + S^{(1)}$

$$= \int_1^2 [2m_H(\tilde{\epsilon} - \tilde{U} - f^{(1)})]^{1/2} dr + \int_1^2 \tilde{P}^2(\tau) d\tau / M, \quad (\text{A.5})$$

$$f^{(1)}(r) = \frac{1}{2M} [\tilde{P}^2(\tau) - \tilde{P}_1^2] - \dot{\tilde{P}}(\tau) \int_1^\tau P(\tau') d\tau' / M. \quad (\text{A.6})$$

As a result the first order corrections $\phi^{(1)}$ and δR will be given by eqs. (43) and (44), and $\delta\tau_H$ in eq. (41) will be

$$\delta\tau_H = d\phi^{(1)}/dE + \tau_H dV(R^*)/dE. \quad (\text{A.7})$$

It appeared that $\delta\tau_H \ll \tau_H$ and $\delta\tau_H$ was neglected in the expression (41) for the temperature. For calculating the pre-exponential factor A_0, B in eqs. (28) and (34) take into account that the variation in the $\text{Im } \phi_1$ at fixed energy E leads to variation in the H-transfer distance R_1

$$\Delta R_1 = P_1 \Delta\tau_1 / M = P_1 \text{Im } \phi_1 / (M_1 \Omega_1). \quad (\text{A.8})$$

Correspondingly the final vibrational energy varies so that

$$\frac{dn_2}{dR_1} = \frac{1}{\Omega_1} \frac{\partial}{\partial R_1} \left(V_a(R_1) - V_2(R_1 + \delta R) - \frac{(P_2^2 - P_1^2)}{2M} \right), \quad (\text{A.9})$$

where $V_2(R)$ is the potential profile in the second channel at $r_{AH} = r^e(R)$. This gives for the pre-exponential factor (33) of the symmetric reaction the following expression

$$A_R^{\text{sym}} = A_0 = \frac{\hbar \Omega^2 M}{\pi} \left[\left(-2 \frac{dV_a}{dR} + \frac{d\epsilon}{dR} \right) \tilde{G} + \frac{d\tilde{G}}{dR} \frac{\tilde{G}^2}{2M} - \tilde{G} \frac{dV_2}{dR} Z \right]^{-1} \quad (\text{A.10})$$

where

$$Z = \delta R \frac{d^2 V_2}{dR^2} \left(\frac{dV_2}{dR} \right)^{-1} + \tau_H \frac{dV_a}{dR} \frac{1}{|P_1|} + \frac{dL}{dR} \frac{1}{M}, \quad (\text{A.11})$$

$$\tilde{G} = \frac{\partial S(R, \epsilon)}{\partial R}; \quad L(R) = \int_1^2 d\tau \int_1^\tau \frac{\partial U}{\partial R} d\tau. \quad (\text{A.12})$$

For exothermic reactions take into account that

$$\frac{d \text{Im } \phi_2}{dR_1} = \Omega_2 \frac{\partial P_2(R_1 E)}{\partial R_1} \left(\frac{dV_2}{dR} \right)^{-1}.$$

Then the pre-exponent $A_0 B$ in eq. (34) may be written as

$$A_R = A_0 B = \frac{\Omega}{2} \left[\frac{\pi}{M} \left(-\frac{dV_a}{dR} \tilde{G} + \frac{d\tilde{G}}{dR} \frac{\tilde{G}^2}{2M} \right) (1 + Z) \right]^{-1/2}. \quad (\text{A.13})$$

It is easy to see that the lowest order of (A.13) and (A.10) in m_H/M with condition $\tau_H = 0$ of the instantaneous H-transfer coincides with (16) and (21) respectively with the replacement $G \rightarrow \tilde{G}$. The factors A_R, A^{sym} calculated according to eqs. (A.13) and (A.10) agree with CSM ones whereas the expressions (16) and (21) with $G \rightarrow \tilde{G}$ give the difference from CSM ones by a factor about 1.5–2.

It is to be noted also that such a trajectory constructed using the variables r_H and $R = R_{AB}$ neglects the nonorthogonality of channel variables r_a, R_a and r_c, R_c . We have obtained the rigorous expansion in m_H/M using the variables

$$\rho = (R_a^2 m_H / M_a + r_a^2)^{1/2};$$

$$\gamma = \arctg[(R_a/r_a)(m_H/M_a)^{1/2}],$$

in which the turning of the R_c, r_c relative to R_a, r_a was taken into account. However even for $m_A = m_B = 15 m_H$ the results of such calculations practically coincide with those using R_{AB}, r_H variables and are not described here.

Appendix B. The relation of three-dimensional (3-D) and linear (1-D) reaction rates

The common ways of calculating the 3-D rate constant W_{3-D} based on the activated complex theory are inapplicable to the tunneling reactions because the H-transfer time τ_H is less than the periods of the bending and other vibrations. In particular, the attempts [27] to use the decreasing of bending frequencies to obtain the low activation energy $E_a(3-D)$, when $E_a(1-D)$ is high, is incorrect. As may be seen from the present study, the high $E_a(1-D)$ in ref. [27] is connected with the repulsive character of the LEPS-type PES at large R .

Make a most rough estimation $W(3-D)$ based on an equation similar to (11)

$$W_0(3-D) = \int 2\pi(R - R_0)^2 dR \sin \theta d\theta \\ \times \psi_{3-D}^2(|R - R_0|)w(R, \theta),$$

where

$$\psi_{3-D} = (\pi^{-1} X_R^{-2}) \psi_{1-D}^2(|R - R_0|),$$

is the probability of the CH_3 radical to be at a distance $R - R_0$ from its equilibrium position with the bending angle $\angle \text{CHC} = \theta$ (X_R is the zero amplitude of the reactant vibrations) and $w(R, \theta)$ is the probability of the H-transfer at fixed R, θ . Consider that at $\theta \neq \theta$ the increasing of the barrier height by about $\Delta U = k_\theta \theta^2/2$ is accompanied by decreasing of the $w(R, \theta)$ according to

$$\Delta \ln w \approx 2\Delta U \partial S(R, \epsilon) / \partial \epsilon = -\tau_H k_\theta \theta^2,$$

where τ_H is the H-transfer time. Then the following estimation may be given

$$W_0(3-D) = \left(\frac{R^*(0) - R_0}{X_R} \right)^2 \frac{\hbar}{\tau_H k_\theta} \approx 1.44 W_0(1-D).$$

The last quantity corresponds to typical values $R^*(0) - R_0 = 0.53 \text{ \AA}$, $X_R = 0.175 \text{ \AA}$, $k_\theta = 10 \text{ kcal/mol rad}^2$ (the PES II.1 with $d = 2.5$).

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